

DETAILED RESEARCH PAPER ON RENEWABLE ELECTRICITY STORAGE SYSTEM BASED ON CONCEPT OF AIRSHIPS

-BY NIKHIL KUSHWAH (STUDENT)
SAMRAT ASHOK TECHNOLOGICAL INSTITUE, VIDISHA (M.P.)
DEPARTMENT OF ELECTRICAL ENGINEERING

1. WHY WE NEED TO STORE THE RENEWABLE ELECTRICITY

Energy Availability During Low Production

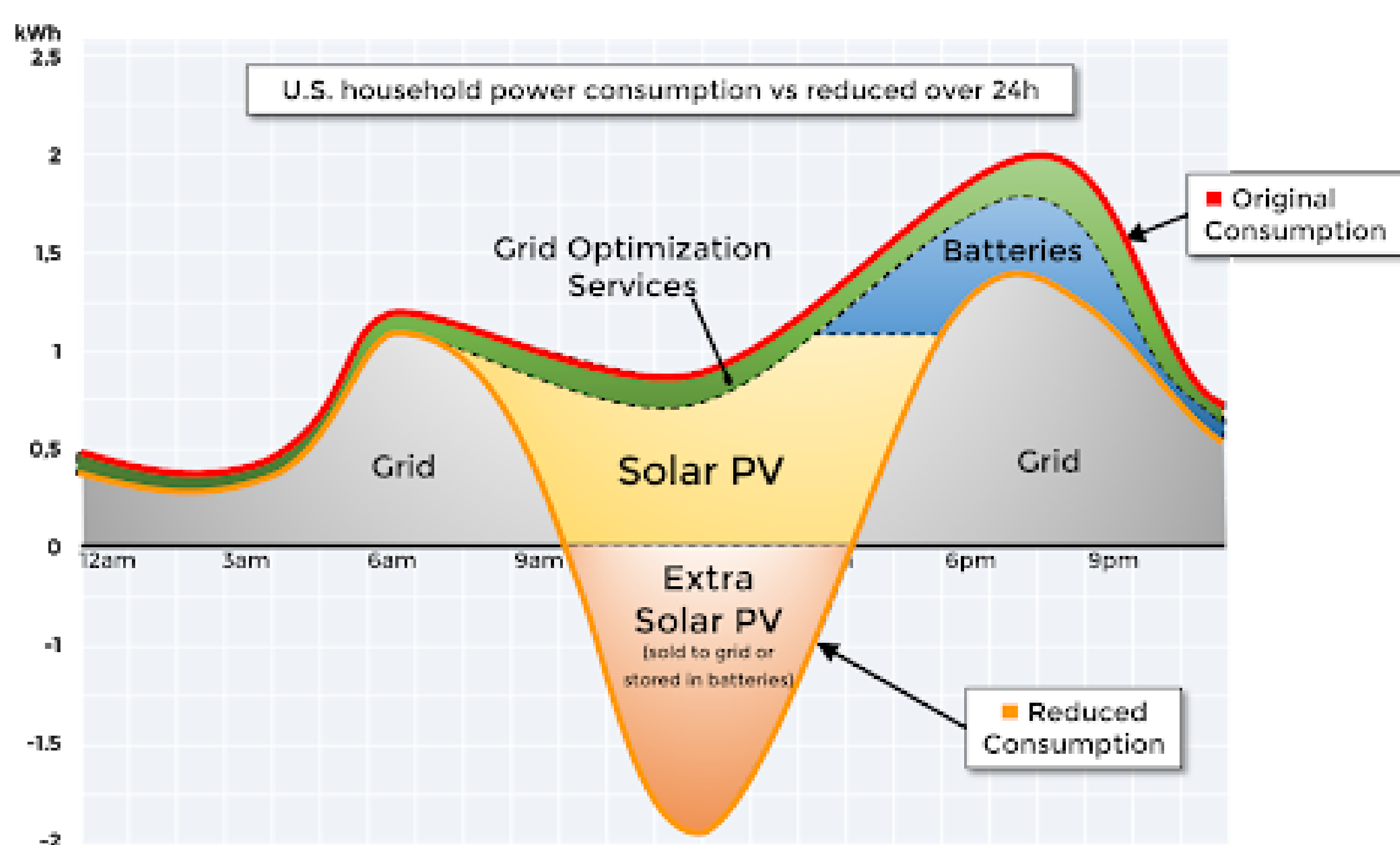
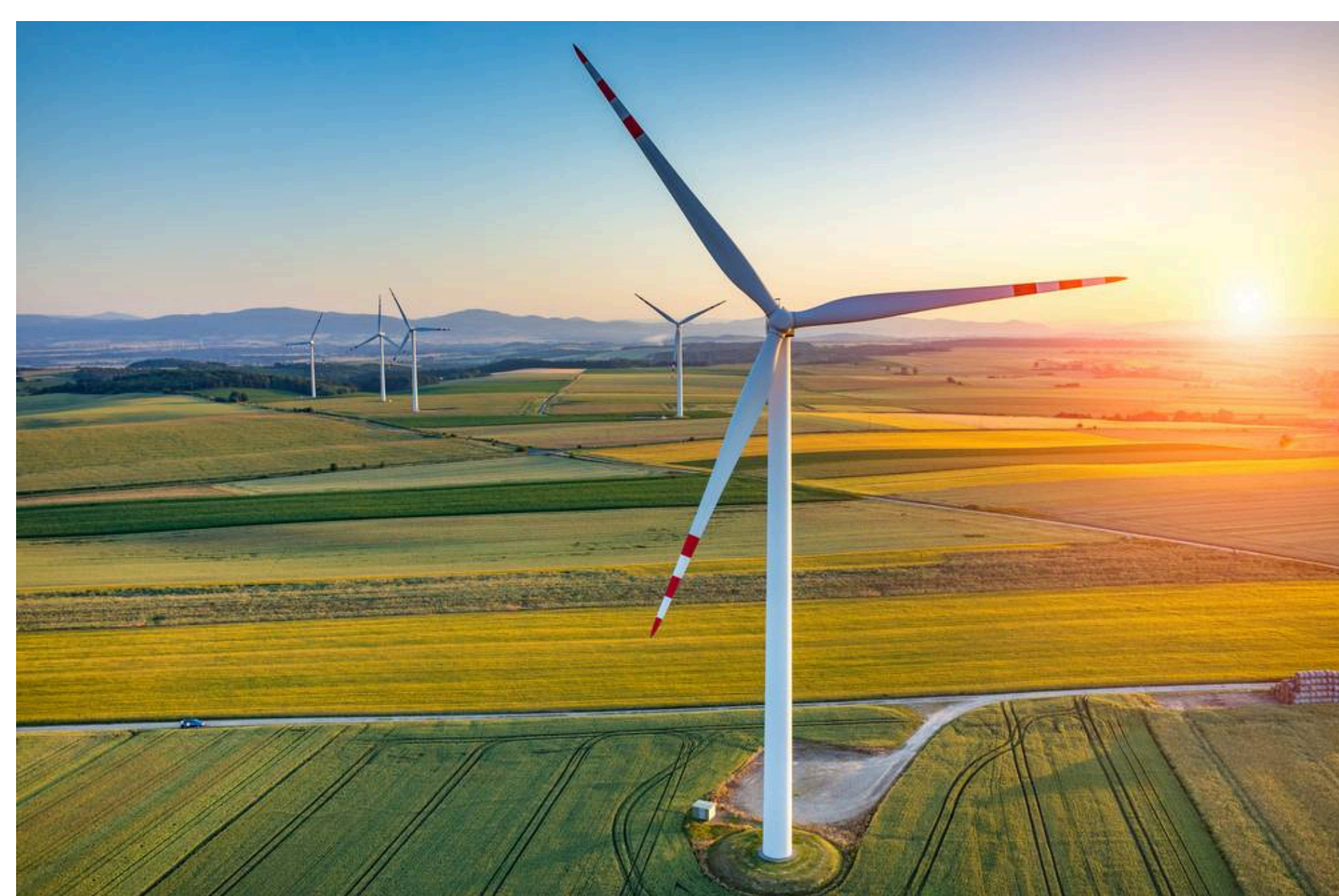
Solar power is only generated during the day, and wind power depends on wind conditions. Energy storage ensures that power is available during times when these sources are not generating energy, like at night

Meeting Demand

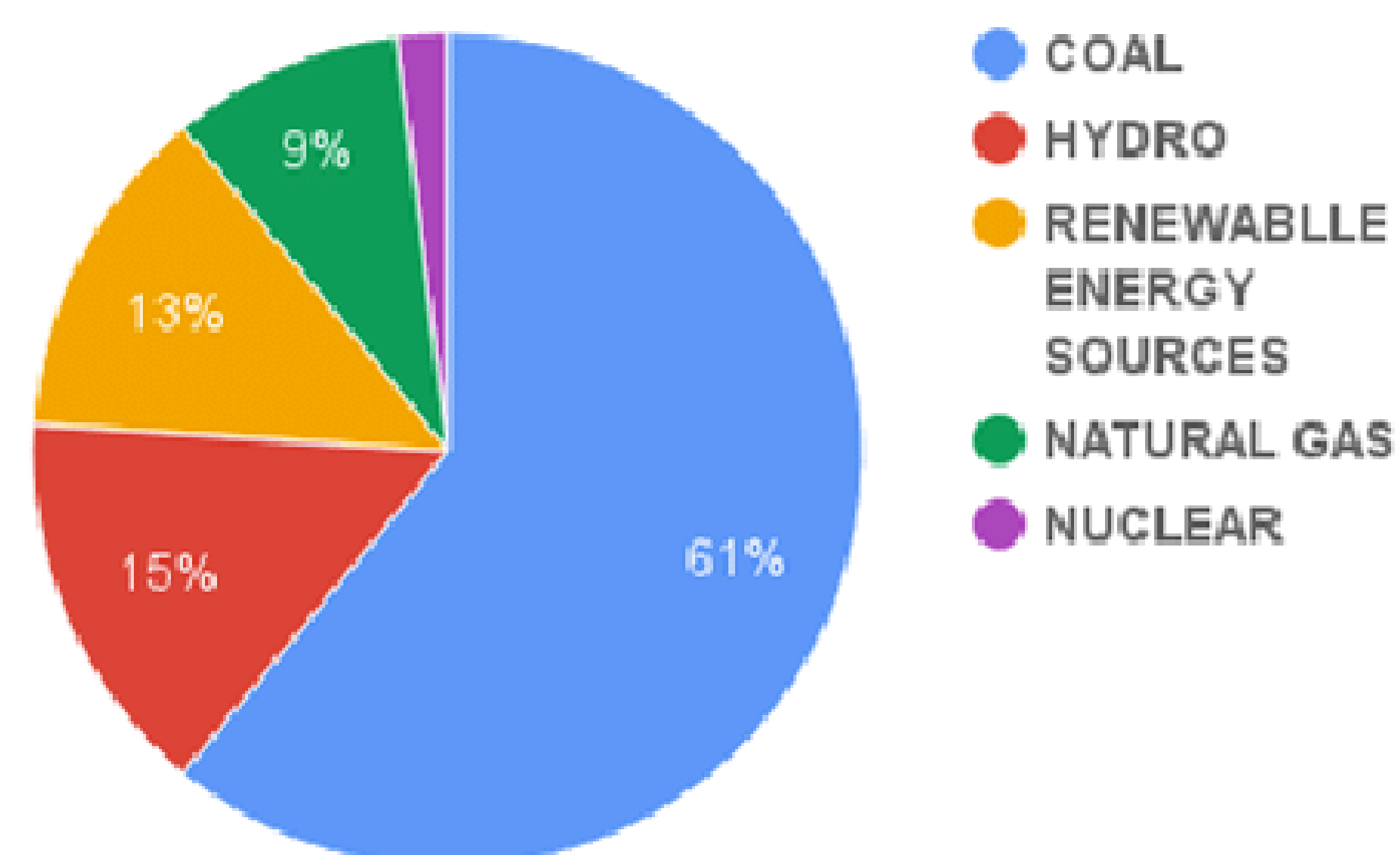
Storage allows for balancing supply and demand. When renewable generation exceeds demand (such as on a sunny day), excess energy can be stored and then used later when demand is higher or generation is lower.

Energy Independence

Storing excess renewable energy improves the overall efficiency of renewable energy systems, reducing waste and making it more cost-effective in the long run.



Sources of Electricity in India By Installed Capacity



Here We Have The Graph Of Extra Solar Energy Produced On day time
(DUCK CURVE)

Contribution OF Different Sources
In Energy Produciton

2. OTHER ENERGY STORAGE SYSTEM AND ITS DISADVANTAGES

2.1. LITHIUM ION

High Initial Cost

Lithium-ion batteries are expensive to manufacture, particularly at grid scale. The cost of raw materials, including lithium, cobalt, and nickel, can be high, making the upfront capital investment substantial.

Limited Lifespan and Degradation

Over time, lithium-ion batteries degrade with each charge and discharge cycle, leading to a reduced capacity and efficiency. This means they need to be replaced periodically, which adds to long-term operational costs.

Thermal Runaway and Safety Risks

Lithium-ion batteries can be prone to overheating and catching fire if not properly managed, especially when used at grid scale.

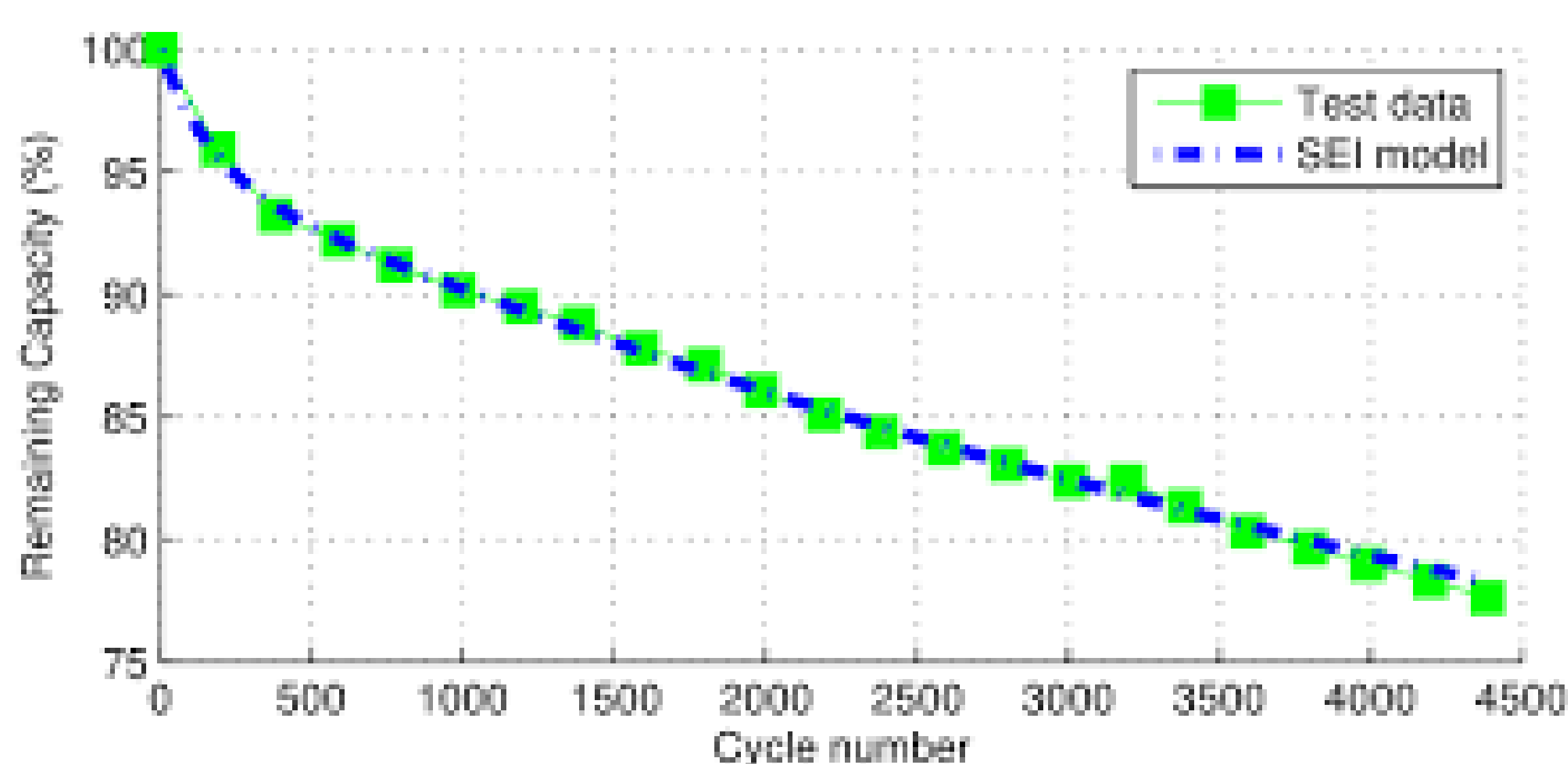
Resource Constraints

Lithium-ion batteries rely on limited raw materials like lithium, cobalt, and nickel, which are finite and often sourced from regions with geopolitical or ethical concerns, such as environmental degradation or human rights abuses in mining.

Environmental Impact

The production, recycling, and disposal of lithium-ion batteries involve chemical processes that can be environmentally damaging. Mining for lithium and cobalt is particularly harmful to ecosystems, and improper disposal of spent batteries can lead to hazardous waste.

Efficiency: 85% to 95%



DEGRADATION OF LI-ION BATTERY

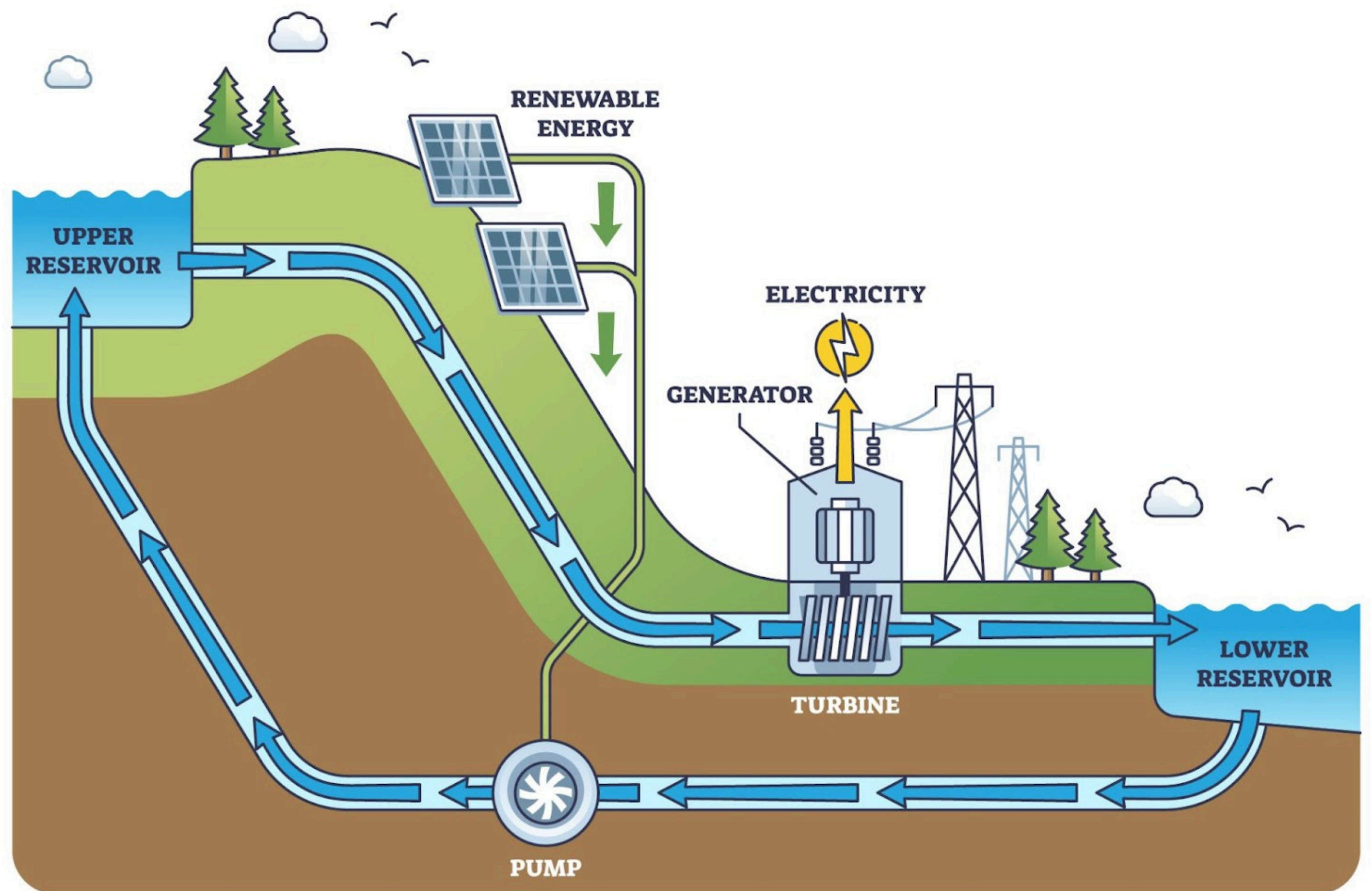


PUMPED HYDROPOWER STORAGE

2.2.PUMPED HYDRO

Geographical Limitations

Pumped hydro requires specific topographical conditions, such as nearby reservoirs at different elevations, to operate. This limits its deployment to areas with suitable geography, and finding appropriate locations can be challenging.



High Capital Costs

The initial cost of constructing a pumped hydro system is very high, especially due to the need for extensive civil engineering work, such as building dams, reservoirs, and tunnels. The long development time and regulatory hurdles further increase costs.

Environmental Impact

The construction of reservoirs and dams can have significant environmental consequences, including disruption of local ecosystems, deforestation, habitat destruction, and changes to river flow.

Water Availability

Pumped hydro relies on large amounts of water, so regions with water scarcity or variable water supply may not be suitable.

Large Space Requirements

The size of the reservoirs needed for pumped hydro systems means that they require large areas of land, which can be a limitation in densely populated or land-scarce regions.

Low Energy Density

Compared to other energy storage technologies, such as batteries, pumped hydro has relatively low energy density, meaning it requires vast amounts of water and elevation to store substantial amounts of energy.

Efficiency: 70% to 85%

2.3 VERTICAL GRAVITY BATTERY

Limited Energy Storage Capacity

The amount of energy stored is directly proportional to the height of the lift and the mass of the weight. In urban areas, the height available for vertical systems is often constrained, limiting the total energy storage.

Infrastructure Costs

Setting up the infrastructure to build tall, durable structures and systems to move large masses vertically can be expensive. This may limit its scalability and competitiveness compared to other energy storage technologies.

Complexity in Scaling

While horizontal systems (like gravity trains) may extend over large distances to increase capacity, vertical systems are restricted by the physical height that can be practically built.

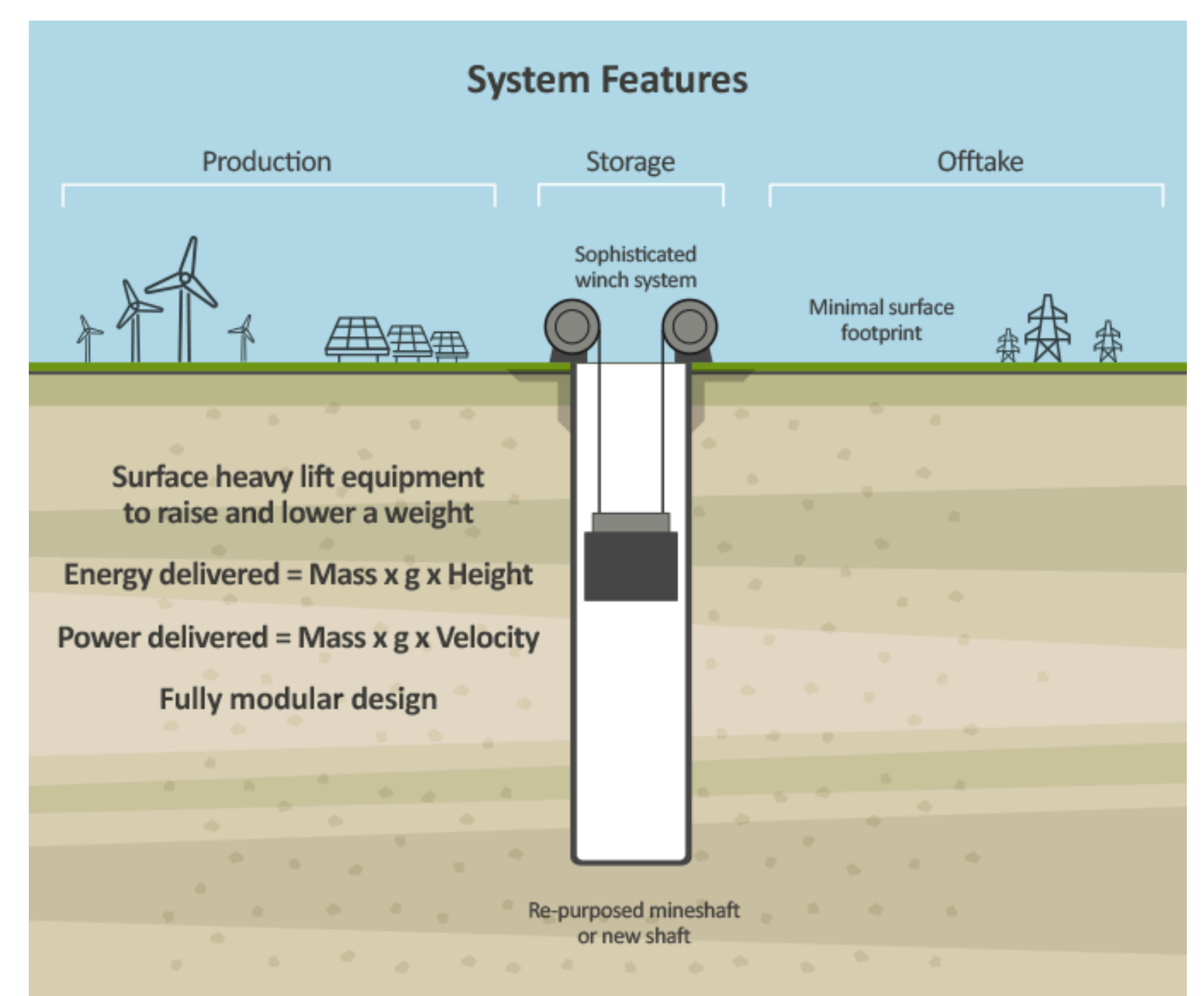
Water Infiltration

Digging deep into the ground could expose the structure to groundwater, leading to water seepage into the system. This may require constant dewatering or waterproofing measures, increasing maintenance costs.

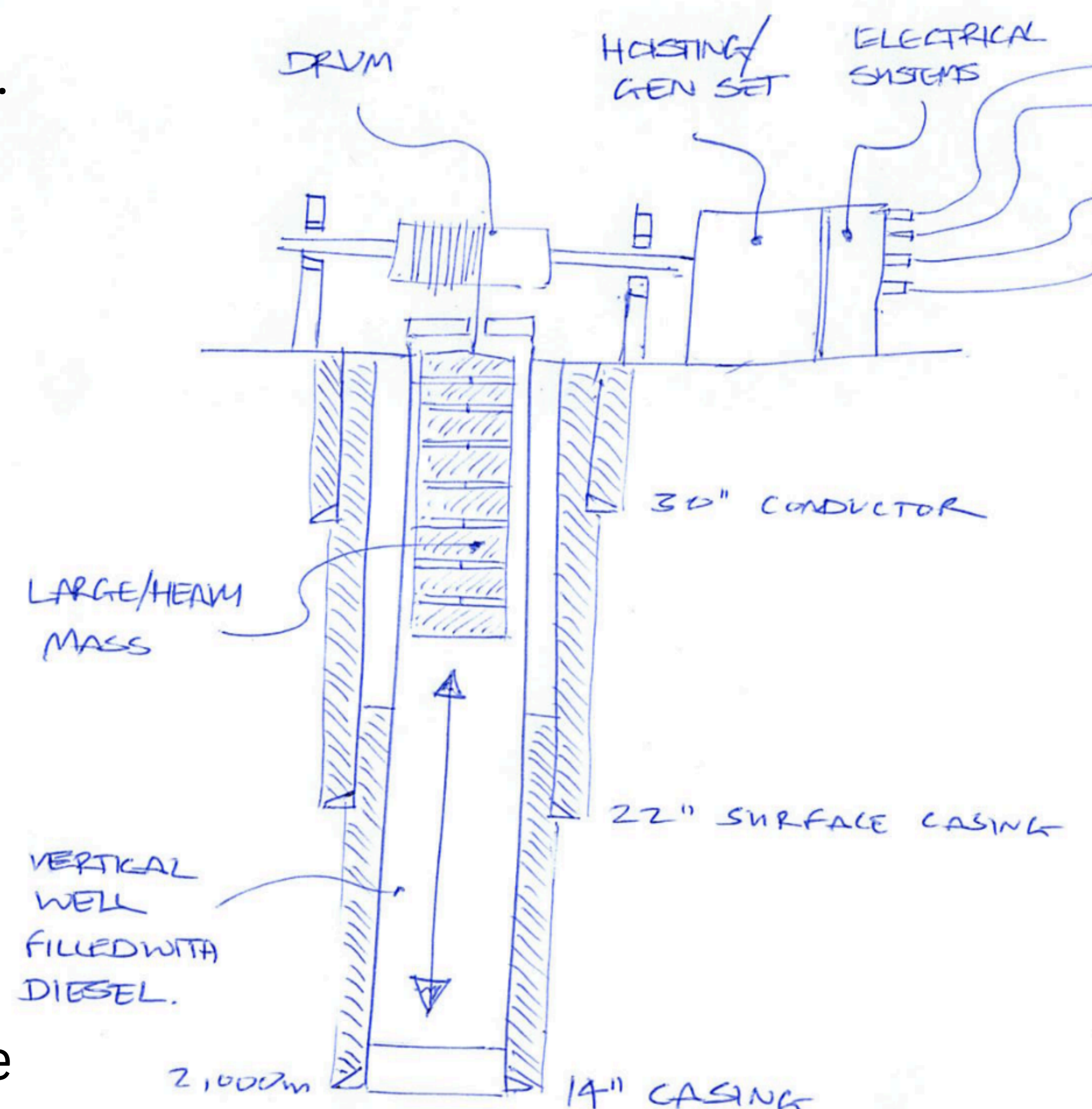
Structural Instability

Underground water can weaken the soil, potentially causing structural instability in the shaft or surrounding infrastructure. This may necessitate additional reinforcement, driving up the cost and complexity.

Efficiency: 60% to 85%

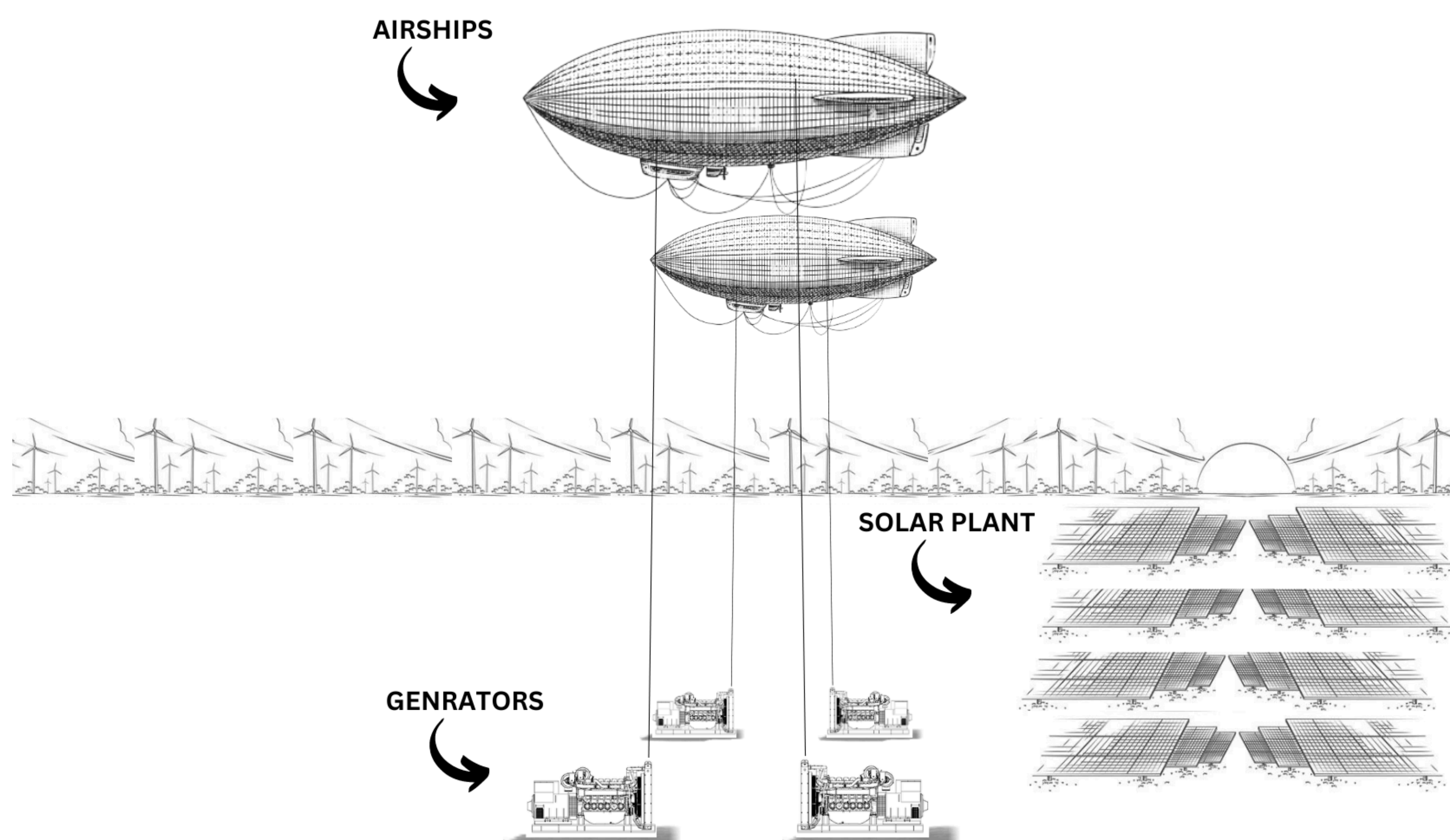


OVERALL DIAGRAM



DETAILED DIAGRAM

AIRSHIP BASED ENERGY STORAGE



1. WORKING OF AIRSHIP BASED BATTERY

Energy Storage (Charging):

When there is excess energy on the grid (e.g., from solar or wind power), the system uses this energy to power electric motors that pull down the airship

the airship store the electricity against air force

Energy Release (Discharging):

When energy is needed, Airship is released in the air at that time the airship is connected with the generators and the generators start moving and convert the stored potential energy to the electrical energy till the airship attain the maximum height

2. ADVANTAGES OF GRAVITY BATTERY

2.1. Scalability

Train gravity batteries can be easily scaled by adjusting the weight of the trains or the length and incline of the tracks. This makes it flexible for different energy storage needs, from small to large-scale applications.

2.2. Long Lifespan

Mechanical systems like airship based energy storage system tend to have long operational lifespans with relatively low maintenance when compared to chemical batteries, which degrade over time. The mechanical nature of the system means that it can last decades with proper care.

2.3. Energy Efficiency

Once the energy is stored in the form of potential energy (pulling down the airship against air), it can be efficiently recovered when the airship release in teh air , converting the stored energy back into electrical energy. With proper motor and generator setups, energy losses can be minimized.

2.4. Energy Storage Over Long Durations

Airship based batteries can store energy for long periods without significant losses, making them suitable for balancing supply and demand over hours, days, or even seasons.

2.5 Simplicity of Operation

The mechanical nature of the system is relatively simple compared to chemical battery systems, reducing the risk of failure and complexity in operation.